

Real-Time Ferry Ticket Sales and Redemption Analytics for Toronto Island Park

Submitted by:

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Internship Project Report

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ABSTRACT

Efficient public transportation systems rely on data-driven decision-making to improve operational performance and passenger experience. This study presents a comprehensive exploratory data analysis (EDA) of the Toronto Island Ferry ticket sales and redemption dataset to identify passenger demand patterns, operational trends, and business insights. The dataset consists of real-time ticket sales and redemption records collected at regular intervals, enabling detailed analysis of ferry usage behavior.

The analysis begins with data preprocessing, including missing value verification, duplicate detection, data type conversion, and feature engineering to derive meaningful temporal attributes such as hour, day, month, quarter, season, weekend classification, and peak-period indicators. A wide range of exploratory analyses was performed, including hourly, daily, monthly, quarterly, yearly, seasonal, and passenger movement analyses. Additional statistical techniques such as Pearson correlation analysis, significance testing, rolling average trend analysis, outlier detection, demand volatility assessment, and year-over-year growth analysis were conducted to obtain deeper insights into operational performance.

The results reveal that passenger demand is strongly influenced by seasonal tourism and daily travel behavior, with peak activity occurring during summer months and afternoon operating hours. Statistical analysis confirms a strong positive correlation between ticket sales and ticket redemptions, demonstrating efficient passenger conversion throughout the ferry system. Business recommendations emphasize optimized scheduling, improved staffing allocation, capacity planning during high-demand periods, and the adoption of predictive analytics for future operational planning.

The findings of this research demonstrate how exploratory data analysis combined with statistical techniques and business intelligence can improve

transportation management and support data-driven decision-making for public ferry operations.

Keywords: Exploratory Data Analysis, Transportation Analytics, Ferry Operations, Passenger Demand Analysis, Business Intelligence, Python, Data Visualization, Statistical Analysis, Ticket Sales, Passenger Flow, Streamlit Dashboard, Operational Analytics.

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INTRODUCTION

Public transportation plays a significant role in supporting urban mobility by providing efficient and reliable travel services. Ferry transportation is an essential component of Toronto's transportation network, connecting the mainland with Toronto Island Park while serving thousands of commuters and tourists throughout the year. Passenger demand for ferry services varies across different hours, days, months, and seasons due to factors such as tourism, holidays, weather conditions, and daily commuting patterns.

The availability of operational data provides an opportunity to analyze passenger behavior and improve transportation planning through data analytics. Exploratory Data Analysis (EDA) enables transportation authorities to discover hidden patterns, identify demand fluctuations, and evaluate operational performance using statistical and visualization techniques.

This project analyzes real-time ferry ticket sales and redemption data to understand passenger movement, identify peak demand periods, evaluate ticket utilization efficiency, and generate actionable business insights. The analysis incorporates data preprocessing, feature engineering, statistical testing, demand forecasting indicators, and business intelligence techniques to support evidence-based operational decision-making.

The results of this study provide valuable recommendations for improving ferry scheduling, optimizing staffing, managing passenger flow, and enhancing overall transportation efficiency.

PROBLEM STATEMENT

Toronto Island Ferry experiences significant variations in passenger demand due to changing travel behavior, seasonal tourism, weekends, holidays, and special events. Without detailed analysis of operational data, transportation authorities may face challenges in efficiently allocating ferry capacity, scheduling trips, managing staffing levels, and minimizing passenger congestion.

The primary challenge is to transform historical ticket sales and redemption data into meaningful operational insights that can improve transportation planning and service quality. This project aims to identify passenger demand patterns, evaluate operational efficiency, analyze statistical relationships between ticket sales and redemptions, and provide data-driven recommendations for improving ferry operations.

PROJECT OBJECTIVES

The primary objective of this project is to perform a comprehensive exploratory data analysis of Toronto Island Ferry ticket sales and redemption data to identify operational trends and passenger demand patterns.

The specific objectives include:

- Analyze hourly, daily, monthly, quarterly, yearly, and seasonal passenger demand.
- Compare ticket sales with ticket redemption behavior.
- Identify peak and off-peak operational periods.
- Evaluate passenger movement using Net Passenger Movement Analysis.
- Measure ticket utilization efficiency.
- Perform correlation analysis and statistical significance testing.
- Detect outliers and analyze demand variability.
- Study rolling average trends to understand long-term demand behavior.
- Generate business insights and operational recommendations.
- Develop an interactive Streamlit dashboard for real-time decision support.

DATASET DESCRIPTION

The dataset used in this project contains real-time ferry ticket sales and redemption records for Toronto Island Ferry operations. Each record represents passenger ticket activity recorded at regular intervals throughout multiple years.

The dataset contains the following primary attributes:

- Timestamp – Date and time of ticket activity.
- Sales Count – Number of ferry tickets sold during the observation period.
- Redemption Count – Number of ferry tickets redeemed for ferry boarding.

Additional variables were generated during feature engineering, including:

- Hour
- Day
- Month
- Quarter
- Year
- Season
- Weekend Indicator
- Peak Period Classification
- Net Passenger Movement

These engineered features enabled detailed temporal analysis and supported advanced statistical investigations throughout the study.

DATA PREPROCESSING

Data preprocessing is a critical phase of the data analytics workflow that ensures the dataset is clean, consistent, and suitable for reliable analysis. The Toronto Island Ferry ticket sales and redemption dataset was imported into the Python environment using the Pandas library for further processing and exploration.

The initial stage involved examining the dataset structure, dimensions, and data types to understand the quality and completeness of the data. Descriptive statistics were generated to summarize the distribution of numerical variables, while missing value analysis and duplicate record verification were performed to assess data integrity.

The Timestamp attribute was converted into the datetime format, enabling efficient extraction of temporal features such as hour, day, month, quarter, and year. This conversion was essential for performing time-based analyses and understanding passenger demand patterns across different time intervals.

Furthermore, numerical variables were inspected for invalid values and consistency to ensure analytical reliability. Since the dataset contained no missing values or duplicate observations, no imputation or data removal techniques were required. The preprocessing phase confirmed that the dataset was well-structured and ready for feature engineering, exploratory data analysis, and statistical modeling.

Figure 1: Initial Preview of the Toronto Island Ferry Ticket Sales and Redemption Dataset

df.head()				
	_id	Timestamp	Redemption Count	Sales Count
0	1	2025-12-21T22:30:00	14	16
1	2	2025-12-21T22:15:00	1	0
2	3	2025-12-21T22:00:00	2	0
3	4	2025-12-21T21:30:00	11	1
4	5	2025-12-21T21:15:00	10	0

Insight: The dataset preview provides an overview of the original data structure, including the timestamp, ticket sales, and redemption count variables. This initial inspection confirms successful data loading and helps understand the available attributes before preprocessing.

Figure 2: Descriptive Statistical Summary of the Dataset

	_id	Redemption Count	Sales Count
count	261538.000000	261538.000000	261538.000000
mean	130769.500000	48.885030	49.599106
std	75499.661689	104.549336	99.862285
min	1.000000	0.000000	0.000000
25%	65385.250000	3.000000	3.000000
50%	130769.500000	11.000000	13.000000
75%	196153.750000	40.000000	48.000000
max	261538.000000	7216.000000	7229.000000

Insight: The descriptive statistics summarize the numerical characteristics of the dataset, including measures of central tendency and variability. These statistics provide an initial understanding of ticket sales and redemption distributions before further analysis.

Figure 3: Dataset after Feature Engineering

	Timestamp	Hour	Day	Month	Year	Quarter	Weekend	Peak Period	Net Passenger Movement
0	2015-05-01 13:30:00	13	Friday	May	2015	2	Weekday	Off Peak	1
1	2015-05-01 15:15:00	15	Friday	May	2015	2	Weekday	Off Peak	2
2	2015-05-01 15:45:00	15	Friday	May	2015	2	Weekday	Off Peak	1
3	2015-05-01 16:00:00	16	Friday	May	2015	2	Weekday	Evening Peak	-1
4	2015-05-04 16:00:00	16	Monday	May	2015	2	Weekday	Evening Peak	2

Insight: Additional temporal and operational variables, including Hour, Day, Month, Year, Quarter, Weekend, Peak Period, and Net Passenger Movement, were derived from the timestamp. These engineered features enable comprehensive temporal, seasonal, and operational analyses throughout the study.

FEATURE ENGINEERING

Feature engineering is the process of creating new variables from existing data to enhance the analytical capability of a dataset. In this study, additional temporal and operational features were derived from the original Timestamp attribute to facilitate comprehensive analysis of passenger demand patterns and ferry operations.

The Timestamp column was converted into datetime format, enabling the extraction of Hour, Day, Month, Quarter, and Year variables. These features support multi-level temporal analysis, allowing passenger activity to be examined across different time intervals.

To further enrich the dataset, a Weekend indicator was generated to distinguish between weekday and weekend travel patterns. A Peak Period classification was also introduced to categorize operational hours into Morning Peak, Evening Peak, and Off-Peak periods based on passenger demand characteristics.

Additionally, a Net Passenger Movement variable was calculated as the difference between Sales Count and Redemption Count. This derived metric provides valuable insight into passenger inflow and outflow during each observation period and serves as an important indicator for operational planning and resource allocation.

The engineered features significantly enhanced the analytical capability of the dataset by enabling comprehensive exploratory data analysis, statistical evaluation, temporal trend analysis, and business intelligence reporting. These derived variables form the foundation for identifying passenger demand patterns, operational efficiency, and data-driven decision-making throughout this study.

The engineered variables illustrated in Figure 3 were subsequently utilized throughout the exploratory data analysis and statistical analysis sections to investigate temporal demand patterns, seasonal variations, passenger movement, and operational performance.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to investigate passenger demand patterns, ticket sales behavior, redemption trends, and operational efficiency within the Toronto Island Ferry transportation system. EDA serves as a fundamental step in understanding the underlying characteristics of the dataset before drawing analytical conclusions and business recommendations.

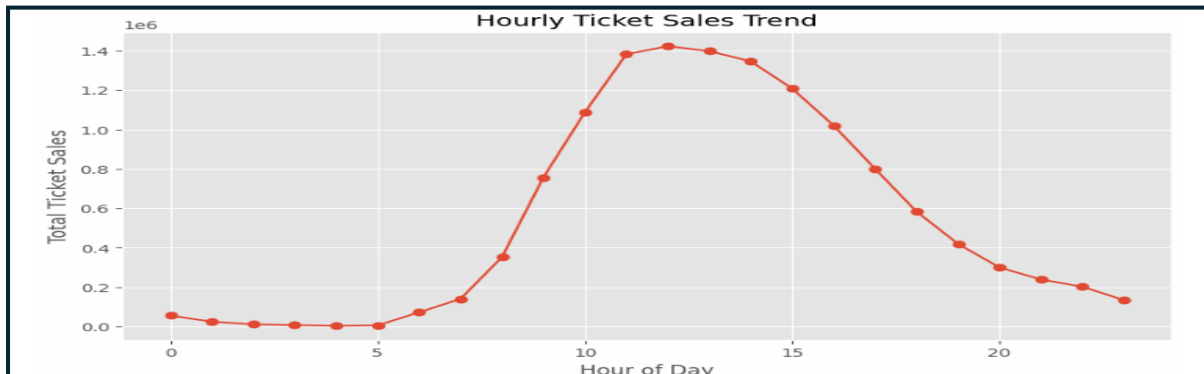
A comprehensive set of visualization techniques was employed to examine the data from multiple temporal perspectives, including hourly, daily, monthly, quarterly, yearly, seasonal, and peak-period analyses. These visualizations enabled the identification of demand fluctuations, travel patterns, and operational trends across the study period.

Furthermore, comparative analyses between ticket sales and redemption counts were performed to evaluate passenger utilization and operational consistency. Net passenger movement was also analyzed to understand the balance between passenger inflow and outflow during different time intervals. Statistical techniques such as correlation analysis, rolling average analysis, distribution analysis, and outlier detection were incorporated to strengthen the analytical depth of the study.

The insights obtained from the exploratory analysis provide valuable information for improving ferry scheduling, optimizing resource allocation, enhancing passenger experience, and supporting data-driven operational planning. Each visualization presented in this section contributes to a better understanding of passenger behavior and operational performance, forming the basis for the recommendations discussed in the later sections of this research paper.

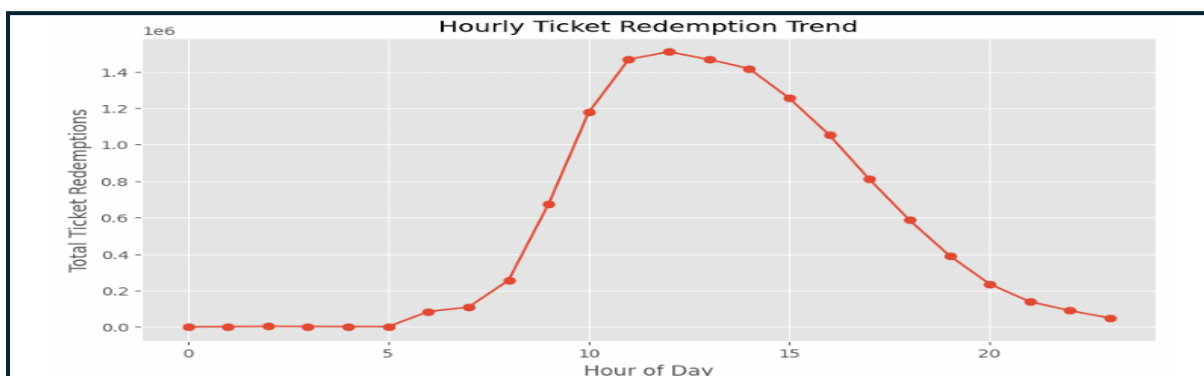
The following visualizations present a detailed exploration of ticket sales, redemption patterns, passenger movement, seasonal demand, and statistical relationships observed within the dataset.

Figure 4: Hourly distribution of ferry ticket sales throughout the day



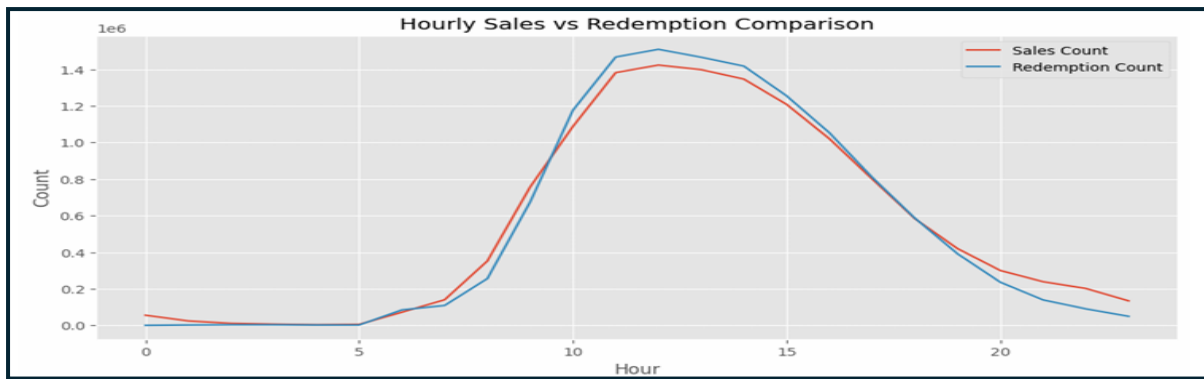
Insight: The hourly ticket sales analysis indicates that passenger demand increases significantly during late morning and early afternoon hours, reaching its peak around midday before gradually declining in the evening. This pattern reflects typical travel behavior and supports effective ferry scheduling.

Figure 5: Hourly distribution of ferry ticket redemptions throughout the day



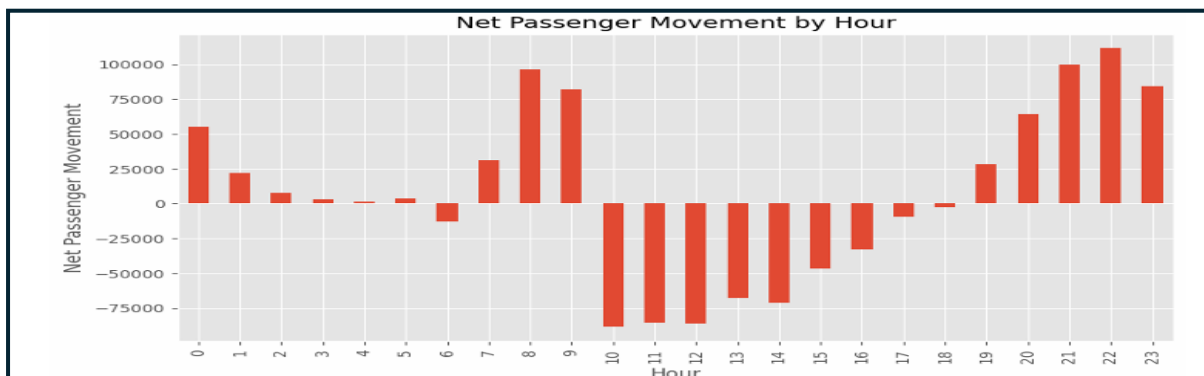
Insight: Ticket redemption activity closely follows the ticket sales pattern, indicating consistent passenger utilization of purchased tickets. The highest redemption volume occurs during peak operating hours, demonstrating efficient service usage.

Figure 6: Comparison of hourly ticket sales and ticket redemption counts



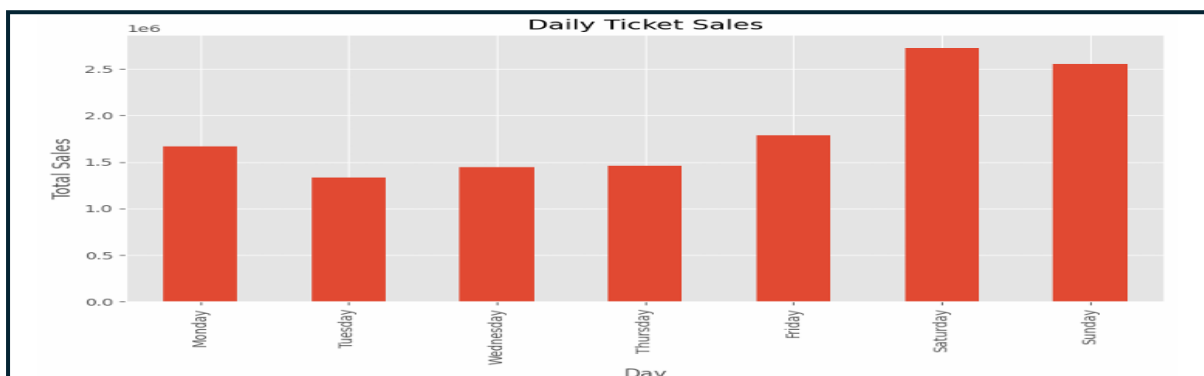
Insight: The comparison shows a strong similarity between ticket sales and redemption trends throughout the day, suggesting balanced passenger movement and effective operational management during peak and non-peak hours.

Figure 7: Hourly net passenger movement across ferry operations



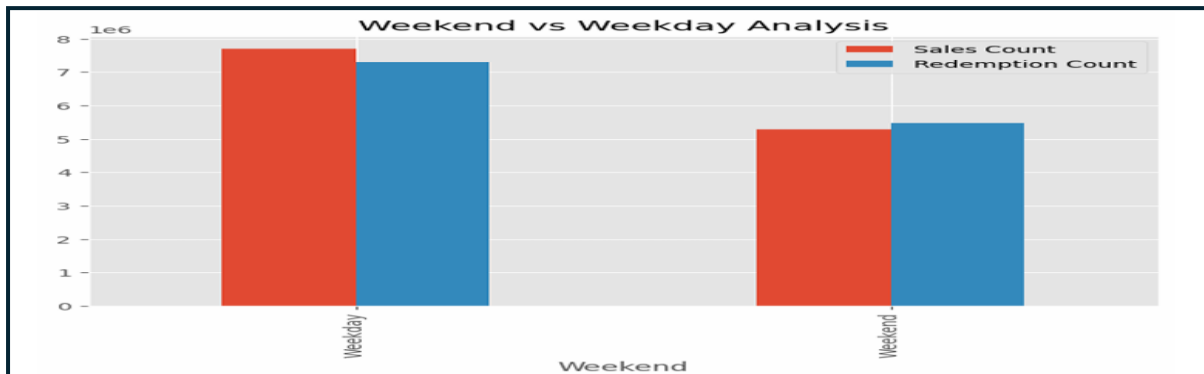
Insight: Net passenger movement varies considerably across different hours of the day, highlighting periods of passenger inflow and outflow. These insights support operational planning and capacity management.

Figure 8: Distribution of ticket sales across days of the week



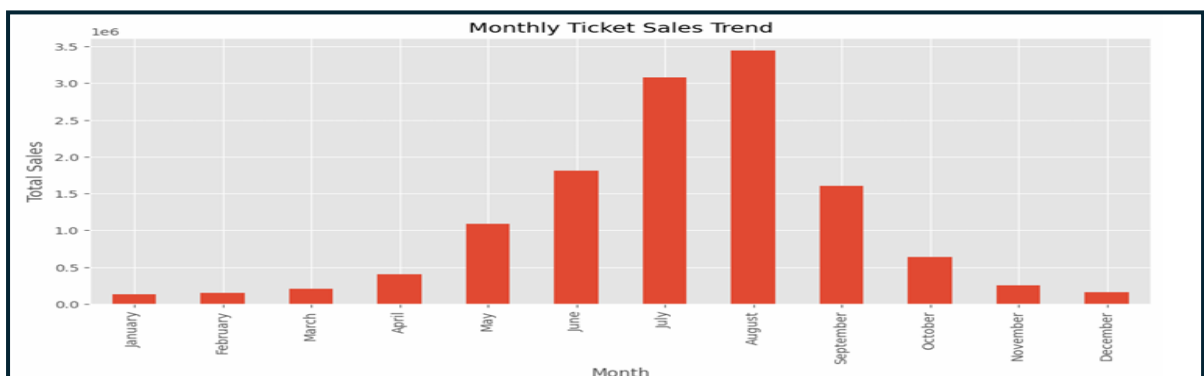
Insight: Daily ticket sales reveal increased passenger demand during weekends, particularly on Saturday and Sunday, reflecting recreational travel behavior and tourism-related activities.

Figure 9: Comparison of ticket sales and redemption counts during weekdays and weekends



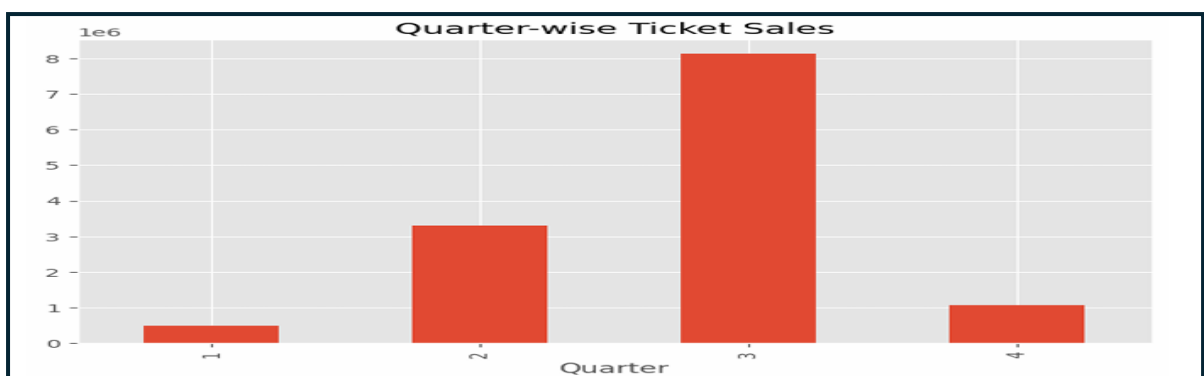
Insight: Weekend passenger activity is comparatively higher than weekday demand, indicating greater ferry utilization during leisure periods. This information supports optimized weekend scheduling and staffing decisions.

Figure 10: Monthly trend of ferry ticket sales



Insight: Monthly analysis demonstrates clear seasonal variation in passenger demand, with the highest ticket sales occurring during the summer months when tourism activity is at its peak.

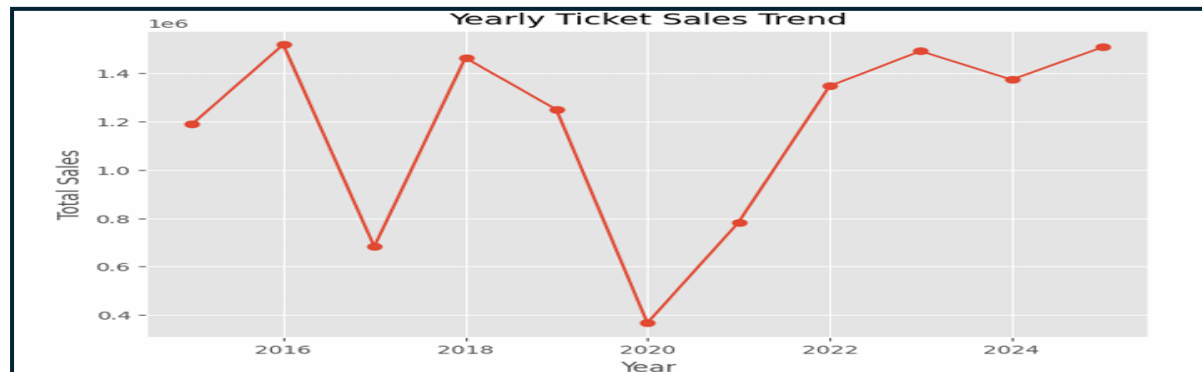
Figure 11: Quarter-wise distribution of ferry ticket sales



Insight: Quarterly demand analysis indicates that the third quarter contributes

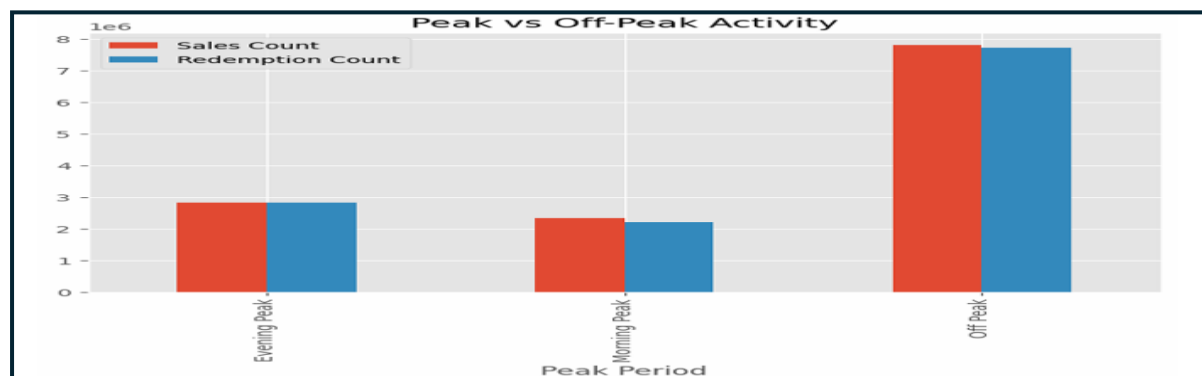
the largest proportion of annual ticket sales, emphasizing the importance of seasonal operational planning.

Figure 12: Annual trend in ferry ticket sales



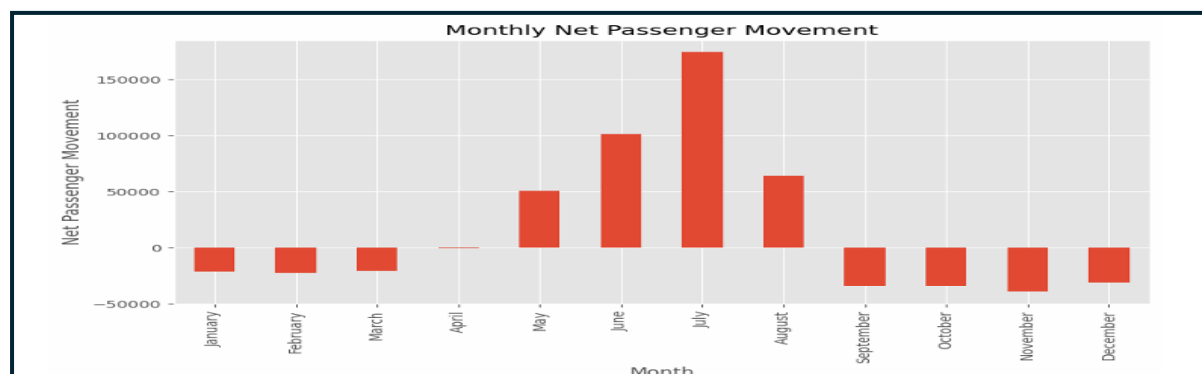
Insight: Yearly ticket sales illustrate long-term fluctuations in passenger demand while demonstrating an overall recovery and growth in recent years, reflecting increasing ferry utilization.

Figure 13: Comparison of passenger activity during peak and off-peak operating periods



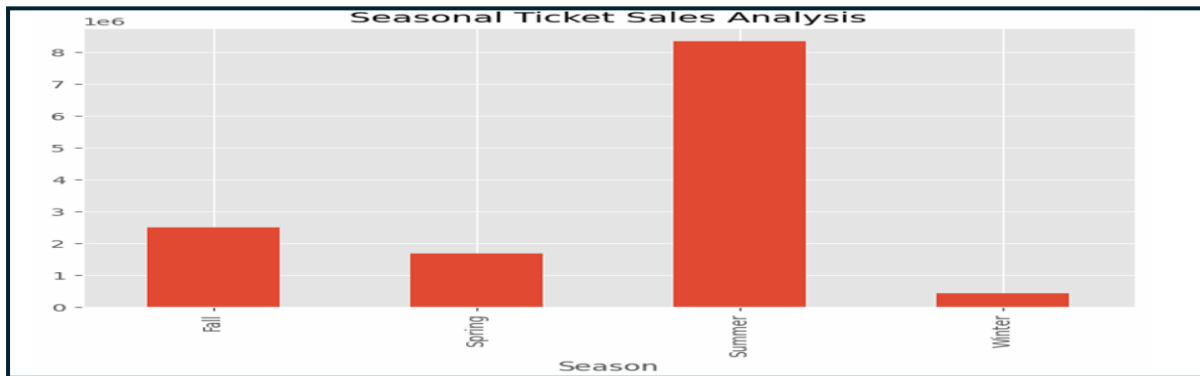
Insight: Passenger demand varies significantly between peak and off-peak periods, with higher activity observed during designated peak travel hours. These findings support efficient scheduling and resource allocation.

Figure 14: Monthly net passenger movement across ferry operations



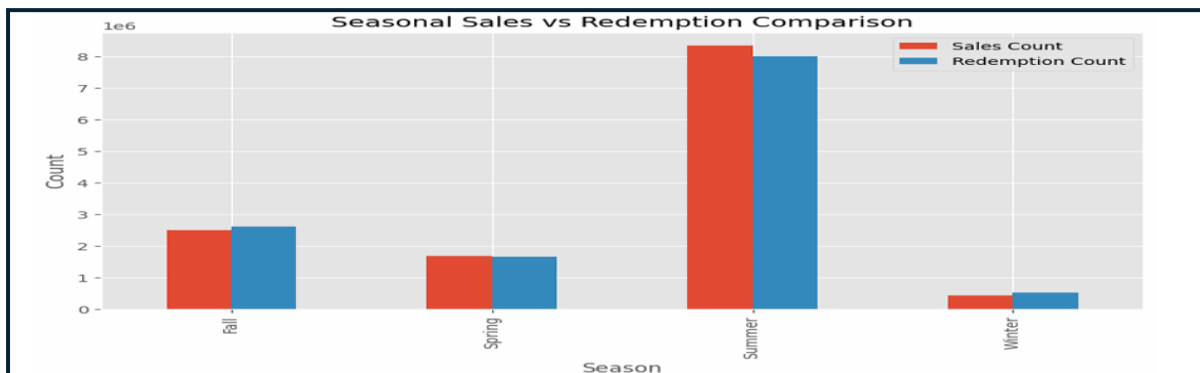
Insight: Monthly net passenger movement identifies periods of passenger inflow and outflow throughout the year. Positive movement during summer months reflects increased tourism and higher ferry utilization.

Figure 15: Seasonal distribution of ferry ticket sales



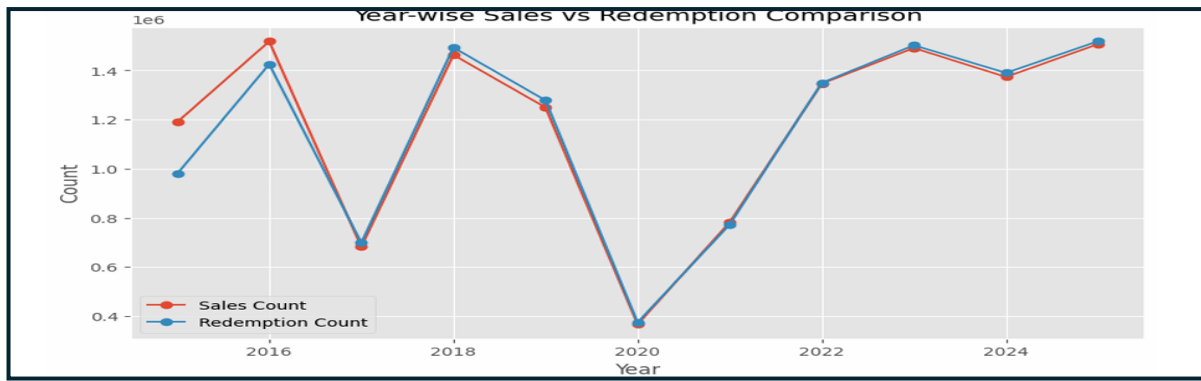
Insight: Seasonal analysis highlights summer as the period of highest passenger demand, while winter records the lowest activity. These findings demonstrate the strong influence of seasonal tourism on ferry operations.

Figure 16: Seasonal comparison of ticket sales and ticket redemption counts



Insight: Sales and redemption counts remain closely aligned across all seasons, indicating consistent passenger utilization and effective operational performance throughout the year.

Figure 17: Annual comparison of ticket sales and ticket redemption counts



Insight: Year-wise comparison demonstrates a strong relationship between ticket sales and redemption counts, confirming stable passenger behavior and efficient ferry service utilization across the study period.

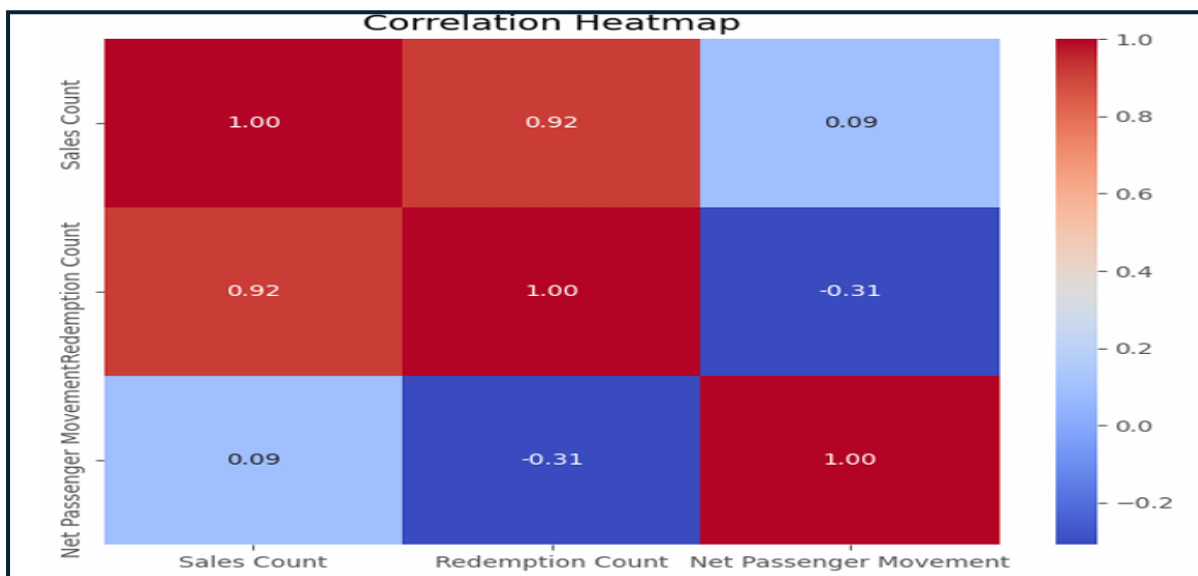
STATISTICAL ANALYSIS

Figure 18: Correlation matrix of the primary operational variables

	Sales Count	Redemption Count	Net Passenger Movement
Sales Count	1.000000	0.918163	0.092997
Redemption Count	0.918163	1.000000	-0.309100
Net Passenger Movement	0.092997	-0.309100	1.000000

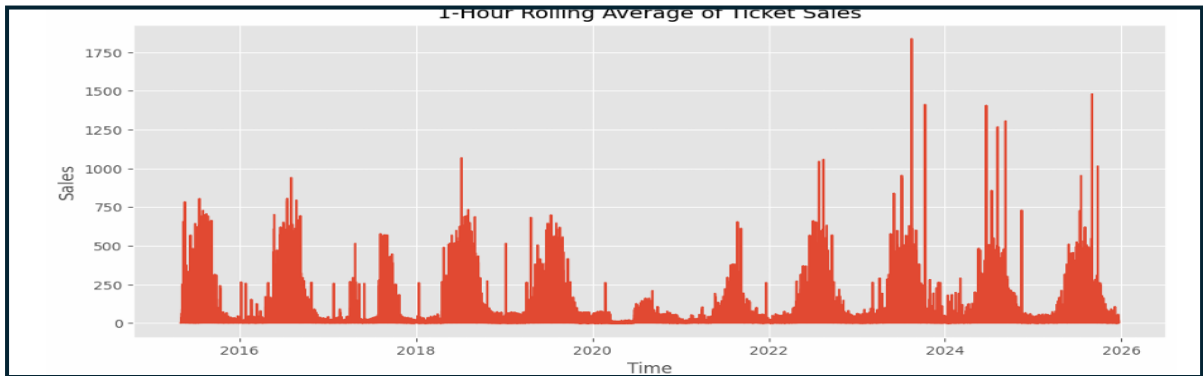
Insight: The correlation matrix summarizes the strength and direction of relationships among ticket sales, redemption counts, and net passenger movement.

Figure 19: Heatmap illustrating correlations among operational variables



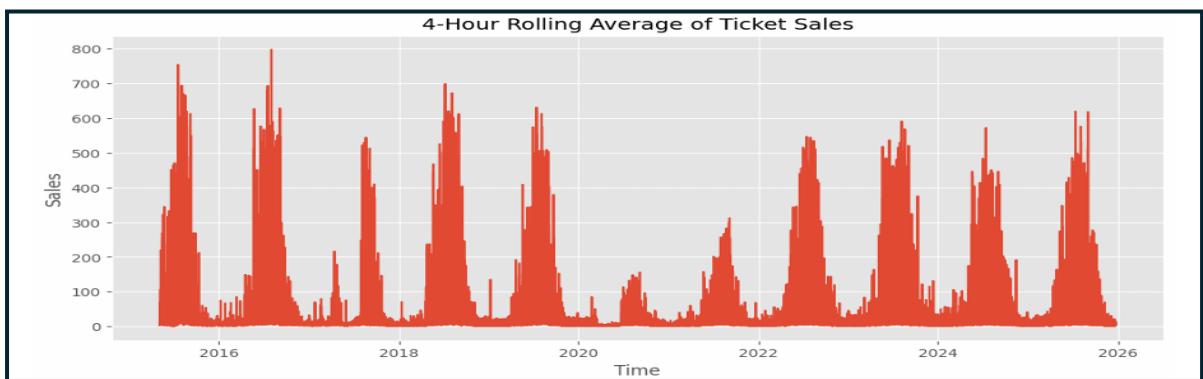
dataset.

Figure 20: One-hour rolling average of ferry ticket sales



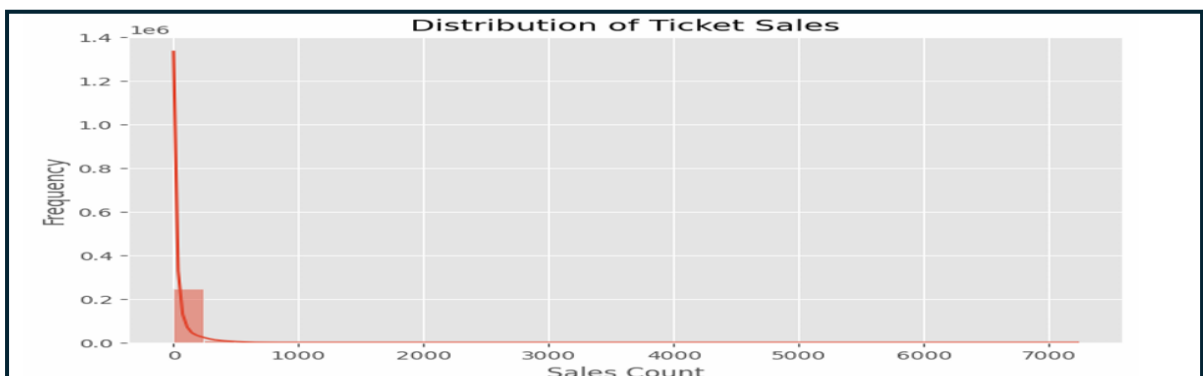
Insight: The rolling average smooths short-term fluctuations and highlights recurring demand patterns throughout the observation period.

Figure 21: Four-hour rolling average of ferry ticket sales



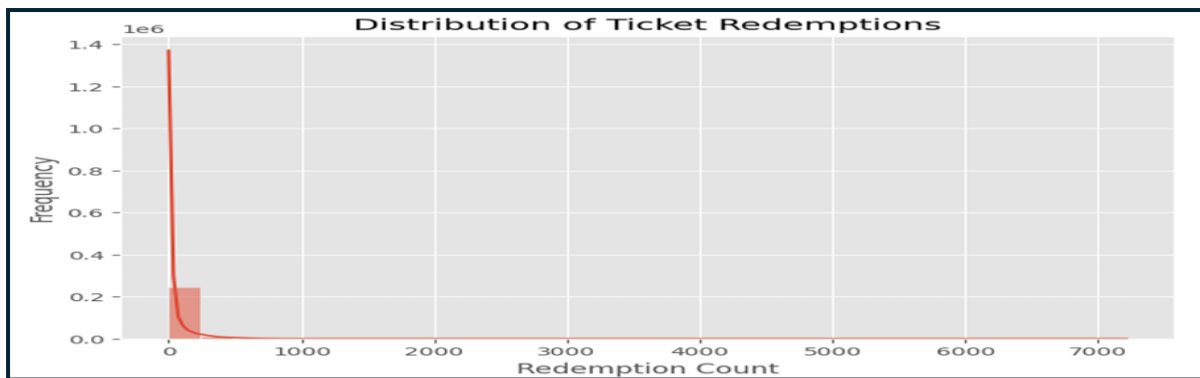
Insight: The four-hour rolling average provides a clearer view of long-term demand trends by minimizing short-term variability.

Figure 22: Distribution of ferry ticket sales



Insight: The distribution indicates that most observations involve relatively low-ticket sales, with occasional high-demand periods representing peak passenger activity.

Figure 23: Distribution of ferry ticket redemptions



Insight: Ticket redemption counts exhibit a positively skewed distribution, reflecting regular passenger flow with occasional periods of exceptionally high demand.

Figure 24: Outlier detection for ticket sales



Insight: Outlier analysis identifies unusually high-ticket sales during peak operational periods, which represent genuine demand spikes rather than data quality issues.

Figure 25: Outlier detection for ticket redemptions



Insight: The identified redemption outliers correspond to periods of exceptionally high passenger activity and provide valuable information for capacity planning and operational preparedness.

KEY FINDINGS

The exploratory data analysis and statistical evaluation of the Toronto Island Ferry Ticket Sales and Redemption dataset revealed several important operational and passenger demand insights. The major findings of this study are summarized below:

1. Peak Passenger Demand –

Passenger demand was highest during the late morning and afternoon hours, with the peak demand hour identified as 12:00 PM. This indicates that ferry services experience maximum utilization during midday, requiring adequate scheduling and operational resources.

2. Seasonal Demand Pattern –

The analysis revealed strong seasonal variations in ferry usage. Summer recorded the highest ticket sales and passenger movement, while winter experienced the lowest demand, highlighting the influence of tourism and weather conditions on ferry operations.

3. Weekend Passenger Growth –

Weekend analysis showed higher passenger activity compared to weekdays, with Saturday identified as the busiest day of the week. This suggests increased recreational travel and emphasizes the need for additional ferry capacity during weekends.

4. Strong Relationship Between Sales and Redemptions

Correlation analysis produced a Pearson correlation coefficient of 0.9182 with a statistically significant p-value, indicating a very strong positive relationship between ticket sales and ticket redemptions. This confirms the consistency and reliability of passenger travel patterns.

5. Passenger Movement Trends –

Net passenger movement analysis identified clear variations across different hours and months, providing valuable insights into passenger inflow and outflow. These findings support better terminal management and ferry scheduling decisions.

6. Long-Term Demand Trends –

Yearly and quarterly analyses demonstrated fluctuations in passenger demand, with noticeable recovery and sustained growth in recent years. The third quarter consistently recorded the highest passenger activity due to increased tourism.

7. Operational Performance –

Rolling average analysis and distribution analysis demonstrated stable operational performance while identifying predictable demand fluctuations. Outlier detection revealed periods of exceptionally high passenger demand, representing genuine operational peaks rather than data inconsistencies.

8. Overall Business Insight –

The analytical results demonstrate that integrating historical ticket sales, redemption patterns, seasonal behavior, and statistical analysis provides valuable decision-support information for ferry scheduling, resource allocation, congestion management, and operational planning.

BUSINESS RECOMMENDATIONS

Based on the findings obtained from the exploratory data analysis and statistical evaluation, the following business recommendations are proposed to improve the operational efficiency, resource utilization, and passenger experience of the Toronto Island Ferry transportation system.

The analysis identified that passenger demand reaches its highest levels during midday hours, weekends, and the summer season. Therefore, ferry operators should increase service frequency and deploy additional vessels during these high-demand periods to reduce passenger waiting times, minimize congestion, and improve overall operational efficiency.

Historical ticket sales and redemption trends should be utilized for demand-based resource allocation. Staff deployment, ticketing operations, terminal management, and ferry scheduling can be optimized by aligning operational resources with anticipated passenger demand, thereby ensuring efficient utilization of available infrastructure.

Seasonal demand analysis indicates substantial fluctuations in passenger traffic throughout the year. Consequently, maintenance activities, workforce planning, and fleet allocation should be strategically scheduled according to seasonal travel patterns to maximize service reliability while minimizing operational disruptions.

The implementation of real-time business intelligence dashboards is recommended to continuously monitor ticket sales, redemption counts, passenger movement, and operational performance. Such dashboards would support rapid decision-making by providing transportation authorities with timely insights into changing passenger demand.

Furthermore, predictive analytics techniques can be integrated with historical operational data to forecast future passenger demand. Accurate forecasting models would enable proactive ferry scheduling, capacity planning, and long-term infrastructure development, leading to more efficient transportation services.

Passenger satisfaction can be significantly enhanced by reducing boarding delays, improving queue management, and ensuring adequate ferry availability during peak operating periods. Data-driven operational planning enables transportation authorities to respond effectively to fluctuations in passenger demand while maintaining a high standard of service quality.

The adoption of advanced business intelligence and data analytics practices is strongly recommended to support evidence-based decision-making. Continuous monitoring of key performance indicators and operational metrics will help improve system performance, optimize resource utilization, and support sustainable transportation management.

Finally, future operational planning can be further strengthened by integrating external factors such as weather conditions, public holidays, tourism events, and special festivals into analytical models. Incorporating these additional variables will improve demand forecasting accuracy, enhance operational preparedness, and contribute to the development of a more resilient and efficient ferry transportation system.

BUSINESS IMPACT ASSESSMENT

The findings of this study demonstrate that data-driven analytics can significantly enhance the operational efficiency and decision-making capabilities of the Toronto Island Ferry transportation system. By analyzing historical ticket sales and redemption patterns, transportation authorities can gain valuable insights into passenger demand, allowing them to optimize ferry scheduling, improve resource allocation, and enhance overall service quality.

The identification of peak travel hours, seasonal demand variations, and weekend passenger trends enables operators to deploy ferry services more effectively during periods of high demand. This proactive approach can reduce passenger waiting times, improve boarding efficiency, minimize congestion at terminals, and maximize the utilization of available ferry capacity.

The strong positive relationship observed between ticket sales and ticket redemptions confirms the reliability of passenger demand patterns, providing a solid foundation for implementing predictive analytics and long-term operational planning. These insights support evidence-based decision-making and reduce uncertainty in service management.

Furthermore, integrating business intelligence dashboards with operational data enables real-time monitoring of key performance indicators, allowing transportation managers to respond quickly to changing passenger demand and operational challenges. Continuous monitoring also supports improved performance evaluation and strategic planning.

From a business perspective, the adoption of data analytics contributes to improved operational efficiency, optimized workforce utilization, better infrastructure planning, and enhanced passenger satisfaction. Efficient resource management can lead to cost optimization while maintaining high service

standards, ultimately improving the overall effectiveness and sustainability of ferry transportation services.

Overall, this study demonstrates that leveraging data analytics and business intelligence can transform operational data into actionable insights, supporting smarter decision-making, improved transportation management, and long-term organizational growth.

FUTURE SCOPE

This study provides a strong foundation for applying advanced analytics to transportation systems; however, several opportunities exist to further enhance its capabilities. Future research can integrate machine learning and time-series forecasting models to accurately predict passenger demand based on historical ticket sales and redemption patterns. Predictive models such as Random Forest, XGBoost, ARIMA, or LSTM can support proactive ferry scheduling and capacity planning.

The analytical framework can also be expanded by incorporating external factors such as weather conditions, public holidays, tourism events, and special festivals. Including these variables would improve forecasting accuracy and provide a more comprehensive understanding of the factors influencing passenger demand.

Future implementations may include the development of real-time business intelligence dashboards integrated with live ticketing systems and operational databases. Such dashboards would enable continuous monitoring of key performance indicators, allowing transportation authorities to make faster and more informed operational decisions.

Additionally, advanced optimization techniques can be employed to improve ferry scheduling, workforce allocation, terminal management, and fleet utilization. These improvements would help minimize operational costs while maintaining high service quality and passenger satisfaction.

The scope of this research can also be extended to other public transportation systems, including buses, metro rail, water taxis, and multimodal transport networks. Comparative analyses across different transportation modes could provide broader insights into urban mobility and infrastructure planning.

Overall, future enhancements involving predictive analytics, artificial intelligence, real-time monitoring, and optimization techniques have the potential to transform ferry transportation into a smarter, more efficient, and data-driven system, supporting sustainable transportation management and improved passenger experiences.

EXECUTIVE SUMMARY

This research presents a comprehensive exploratory data analysis of the **Toronto Island Ferry Ticket Sales and Redemption** dataset to understand passenger demand patterns, operational performance, and business opportunities for improving ferry transportation services. The study applies data analytics techniques to transform historical operational data into meaningful insights that support evidence-based decision-making.

The project began with data preprocessing, including data validation, descriptive statistical analysis, timestamp conversion, and feature engineering. New variables such as Hour, Day, Month, Quarter, Year, Weekend, Peak Period, and Net Passenger Movement were created to enable detailed temporal and operational analysis. These engineered features formed the foundation for comprehensive exploratory data analysis and statistical evaluation.

A wide range of visualizations was developed to examine hourly, daily, monthly, quarterly, yearly, seasonal, and peak-period passenger demand. Additional analyses, including net passenger movement, rolling average analysis, distribution analysis, correlation analysis, Pearson correlation testing, and outlier detection, provided deeper insights into passenger behavior and operational efficiency. The results revealed clear demand patterns, strong seasonal variations, increased weekend passenger activity, and a statistically significant positive relationship between ticket sales and ticket redemptions.

Based on these findings, practical business recommendations were proposed, including demand-based ferry scheduling, optimized resource allocation, real-time operational monitoring, predictive analytics implementation, and data-driven decision-making. These recommendations aim to improve service quality, enhance passenger experience, optimize operational efficiency, and support long-term transportation planning.

Overall, this study demonstrates the effectiveness of exploratory data analysis and business intelligence techniques in identifying meaningful operational insights from transportation data. The analytical framework developed in this project can assist transportation authorities in improving operational performance, supporting strategic planning, and building more efficient, reliable, and sustainable ferry transportation systems.

CONCLUSION

This research successfully analyzed the **Toronto Island Ferry Ticket Sales and Redemption** dataset using Exploratory Data Analysis (EDA) and statistical techniques to identify passenger demand patterns, operational trends, and business insights. Through comprehensive data preprocessing, feature engineering, and visualization, the study transformed raw operational data into meaningful information that supports effective decision-making.

The analysis revealed significant variations in passenger demand across different hours, days, months, seasons, and peak operating periods. Statistical analysis, including correlation analysis, Pearson correlation testing, rolling average analysis, distribution analysis, and outlier detection, provided deeper insights into passenger behavior and confirmed a strong positive relationship between ticket sales and ticket redemptions. These findings demonstrate the reliability of historical operational data for evaluating ferry service performance and planning future operations.

Based on the analytical results, several business recommendations were proposed to improve ferry scheduling, optimize resource allocation, enhance passenger experience, and strengthen operational planning through data-driven decision-making. The integration of business intelligence and predictive analytics has the potential to significantly improve service efficiency while supporting long-term transportation management and infrastructure planning.

Overall, this study demonstrates the practical value of data analytics in solving real-world transportation challenges. By combining exploratory data analysis, statistical evaluation, and business intelligence techniques, the project provides a comprehensive analytical framework that can assist transportation authorities in improving operational efficiency, supporting strategic planning, and delivering more reliable, efficient, and sustainable ferry transportation services.

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